

Medical Insurance Charges Analysis

Introduction

This project analyses a medical insurance dataset to understand how patient characteristics relate to insurance charges and to develop a simple predictive model that can support more consistent cost estimation. The analysis considers age, sex, BMI, number of children, smoking status, and region.

Exploratory Data Analysis (EDA), SQL, and Linear Regression were used to identify important patterns, evaluate relationships between patient characteristics and charges, and translate the findings into useful business insights.

Data Preparation and Exploratory Analysis

The original dataset contained 1,338 observations and seven columns. No missing values were identified. One exact duplicate was detected and removed during data cleaning, leaving 1,337 observations for the Python analysis.

Insurance charges were right-skewed, with most observations concentrated among lower charge values and fewer observations at substantially higher values. Age showed a positive relationship with insurance charges, while BMI showed a weaker and less clearly defined relationship when examined independently. Average charges across the number-of-children categories also did not follow a consistent increasing pattern.

Smoking status produced one of the clearest differences in the dataset. Smokers generally had substantially higher insurance charges than non-smokers. However, considerable variation also existed within the smoker category, indicating that smoking status alone does not explain the cost of every patient. Differences across sex and region were less pronounced than the difference observed between smokers and non-smokers.

Potential outliers were investigated rather than automatically removed. Some charges that appeared unusually high within the overall dataset were found to be reasonable when examined within the smoker category. This demonstrated that outlier detection across the entire dataset can produce a different conclusion from investigating observations within relevant groups. Examining the context of these observations helped avoid unnecessary removal of valid data.

Predictive Modelling and Performance

The `charges` column was selected as the dependent variable, while the remaining patient characteristics were used as independent variables. The dataset was divided into 80% training data and 20% testing data.

Sex and smoking status were represented as binary variables, while region was one-hot encoded to prevent the model from interpreting the regional categories as having a mathematical order. Northeast served as the reference region.

A Linear Regression model was trained on the training data and evaluated on the unseen test data. The model achieved an **R² score of approximately 0.807**, meaning that it explained about **80.7% of the variation in insurance charges within the test data**. The remaining variation was not explained by the fitted model.

Key Model Findings

Smoking status showed a particularly strong association with predicted insurance charges. Holding the other included patient characteristics constant, smokers were predicted to have charges approximately **\$23,077.76 higher than non-smokers**.

Age also showed a positive association with charges. Each additional year of age was associated with approximately **\$248.21 higher predicted charges**, while each one-unit increase in BMI was associated with approximately **\$318.70 higher predicted charges**.

Each additional child was associated with approximately **\$533.01 higher predicted charges**. However, during EDA, raw average charges did not increase consistently across the children categories. This highlights the difference between examining group averages independently and estimating a variable's relationship with charges while holding other characteristics constant.

Male patients were predicted to have charges approximately **\$101.54 lower than female patients**. Compared with Northeast, predicted charges were approximately **\$391.76 lower for Northwest, \$838.92 lower for Southeast, and \$659.14 lower for Southwest**. These estimated differences were comparatively modest relative to the difference associated with smoking status.

SQL Findings

SQL analysis provided additional summaries of the dataset. Smokers had an average insurance charge of approximately **\$32,050.23**, compared with **\$8,434.27** for non-smokers. The average charge for smokers was therefore approximately **3.8 times** that of non-smokers.

Average insurance charges also increased across the defined age groups. Patients aged **60 and above had the highest average charge at approximately \$21,248.02**, while patients aged 18–29 had the lowest.

Across BMI categories, the obesity category had the highest average charge at approximately **\$15,552.34**, while the underweight category had the lowest at approximately **\$8,852.20**.

Regional analysis showed that Southeast had the highest raw average charge at approximately **\$14,735.41**, while Southwest had the lowest at approximately **\$12,346.94**. The regional differences were considerably smaller than the difference observed between smokers and non-smokers.

Business Recommendations

The insurance provider should consider multiple patient characteristics together when estimating insurance charges rather than relying on a single characteristic. The analysis shows that several patient attributes contribute useful information, although their relationships with charges differ.

Smoking status should receive appropriate consideration in charge estimation because it showed a strong and consistent association with higher insurance charges across EDA, SQL analysis, and the Linear Regression model. However, smoking status should be considered alongside other patient characteristics because considerable variation exists even among smokers.

The provider should also combine predictive modelling with continued exploratory analysis of patient data. Model coefficients provide useful estimates after accounting for other included variables, while exploratory analysis can reveal distributions, group differences and unusual observations that may not be fully represented by a single model score.

Conclusion

The analysis shows that medical insurance charges are associated with several patient characteristics, with smoking status showing one of the clearest relationships with higher charges. Age, BMI, number of children, sex, and region also contributed information to the Linear Regression model.

The model explained approximately **80.7% of the variation in insurance charges in the test data**, indicating that the included patient characteristics provide substantial information for estimating charges while some variation remains unexplained.

Overall, combining exploratory analysis, SQL summaries, and predictive modelling provides a more complete understanding of insurance charges and can support more consistent, data-informed cost estimation.